

BrickAtlas: Auditing Display-Grounded Evidence Following with Paired Observations of LDraw Parts

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Abstract

001 *Answering a visual question correctly does not establish*
002 *that a model tracks the evidence selected by its reference. A*
003 *paired test can address this gap only if the reference and the*
004 *answer remain well defined after the intervention. We intro-*
005 *duce a display-grounded observation contract on 24 human-*
006 *authored LDraw assemblies. Numeric labels select objects*
007 *in the current image; canonical source identities provide*
008 *audit provenance, not immutable answers to edited images.*
009 *Two separate families test rendered color changes at fixed*
010 *geometry (73 pairs) and candidate-reference reassignment*
011 *for geometry matching (67 pairs). Every exact observation*
012 *is bound to independent human review, and family-specific*
013 *paired correctness is reconstructed from raw responses. A*
014 *supporting graph audit demonstrates why the choice of con-*
015 *trol matters: an isolates-plus-one heuristic solves 72/73 nat-*
016 *ural questions and all 61 degree-visible change pairs, but*
017 *has zero both-correct on 12 degree-matched pairs; BFS*
018 *solves all of them. The strong subset is a small proof of con-*
019 *cept, including an anonymous 1-WL blind spot. We release*
020 *machine-verified stimuli, the executable evaluation proto-*
021 *col, and a searchable 617-item historical dossier. Human*
022 *visual certification and learned-model diagnostic value re-*
023 *main empirical requirements; the reported findings concern*
024 *stimulus and algorithmic audits.*

025 1. Introduction

026 A label can identify a part in a source model, or it can select
027 a visible object in an image. These meanings coincide in an
028 unchanged rendering but can diverge under an intervention.
029 If a different part is relabeled B0035, the original B0035
030 does not acquire a different type or color. A benchmark
031 that changes its gold without specifying which referent the
032 question selects risks rewarding a change in interpretation
033 rather than evidence following.

034 We ask a narrower, observable question: *does a response*
035 *correctly follow the property or candidate reference se-*

lected in the current display, when that observation changes 036
under an explicit contract? 037 The contract resolves numeric
038 references within the supplied image. The source identity
039 is retained for construction, pairing and inspection. The
040 model is never asked to recover a historical source fact from
041 a deliberately edited display.

042 We study two atomic visual families separately (Fig-
043 ure 1). In Color, the same original geometry and pose are
044 rendered with two opaque body colors, while the camera, la-
045 bels and material settings are fixed. The question explicitly
046 concerns the current rendered appearance. In Part-type, tar-
047 get geometry and its label are fixed, while two candidate la-
048 bels exchange their visible referents. The correct candidate
049 token changes; no canonical part changes its type. These in-
050 terventions support different interpretations, so they never
051 contribute to a pooled visual score.

052 Paired image-dependent answers have substantial prece-
053 dent in VQA [3], contrast sets [2], and joint grounding
054 tests [12]. Our contribution is not the invention of pairing
055 or both-correct scoring. It is a versioned measurement con-
056 tract that connects a well-defined current referent, condition-
057 specific gold, independent review of both exact observa-
058 tions, and receipt-derived response transitions in a reusable
059 assembly resource. This connection matters because correct
060 gold in a CAD record does not guarantee a uniquely visible
061 answer.

062 The supporting graph audit gives a concrete example of
063 how inadequate controls can defeat the intended interpre-
064 tation. A shallow rule that counts isolated surviving nodes
065 and adds one for any remaining nodes answers 72/73 natural
066 component questions correctly. It also solves both sides of
067 all 61 change pairs whose degree distributions differ. On 12
068 pairs matched in complete degree sequence, the same rule
069 has zero paired both-correct, while component algorithms
070 solve all pairs. This establishes an interpretive distinction
071 between the distributions. It does not establish that a learned
072 model uses the heuristic.

073 The release contributes (1) two explicit and separately
074 scored display interventions on preserved source geometry;
075 (2) an exact-observation, dual-review protocol and failure-

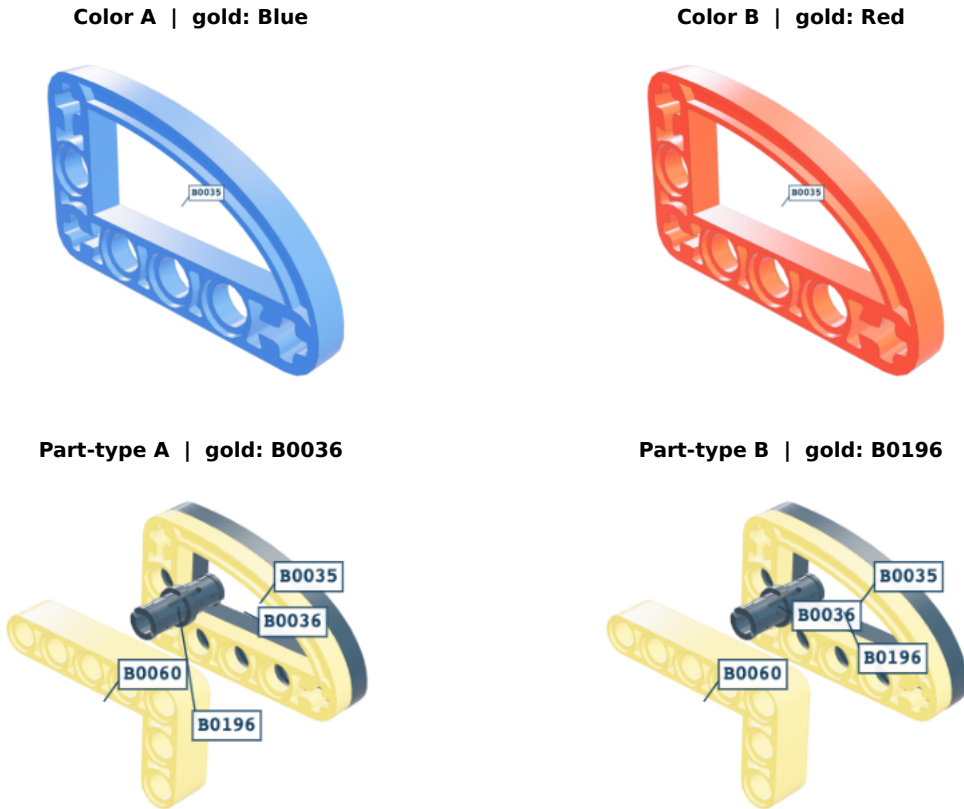


Figure 1. Actual paired observations from OMR 42004, shown with identical within-family paper crops. Top: Color asks for the current color of B0035; only the rendered body-color parameter changes, from Blue to Red. The original source color remains Black in provenance. Bottom: Part-type keeps target B0035 fixed and swaps candidate references B0036 and B0196. The matching geometry is unchanged; the answer changes from B0036 to B0196. Bottom details omit the remote candidate B0116. Gold annotations and crops belong only to this figure. Model inputs are the full 1280×800 PNGs and new display-grounded questions. These are stimulus examples, not model response cases or human answerability judgments.

preserving statistical implementation; and (3) a partitioned graph audit with shallow-feature baselines and transparent limits. We make implementation acceptance distinct from empirical acceptance. The stimuli and algorithms can be checked now; whether the protocol reveals stable learned-model differences beyond ordinary accuracy requires human judgments and genuine model responses.

2. Related Work and Measurement Scope

Assembly benchmarks. BrickNet [6] represents placed brick instances and connectors in graph-backed generative assembly programs. LEGO-Puzzles [11] evaluates spatial and sequential reasoning in editable rendered scenes; its pinned v3 protocol includes three-annotator validation. Phy-Block [8] studies physical understanding and planning with dependencies and simulation. These resources already provide useful identity, rendering and verification machinery. We do not claim that numeric IDs or reusable scene assets

are new. Our visual endpoint is narrower than assembly planning: interpreting a current reference and answering two elementary visual questions under a declared intervention.

Contrastive and consistent answers. Contrast sets probe local decision boundaries using small meaningful input changes [2]. VQA v2 pairs similar images that support different answers to the same question, reducing the utility of language-only shortcuts [3]. Cycle-consistency and VQA-Rephrasings address answer stability under linguistic variation [10]. Required answer change and required invariance are different tests: a constant predictor can be stable without responding to evidence. BrickAtlas records which property an intervention changes and which answer policy applies, rather than treating every response change as either success or failure.

Reference grounding and construct validity. Referring-expression comprehension resolves a phrase against objects and alternatives in an image [9]. Numeric labels form a much narrower interface. Winoground tests joint image-caption matching under controlled word inventories [12]; its paired nature does not make our two atomic attributes an equally broad compositional evaluation. Analyses of Winoground show that small objects, resolution and requirements beyond the nominal construct can explain failures [1]. We therefore require independent judgments of the exact question and pixels, including the uniqueness of a visual answer.

Controlled visual and graph evidence. CLEVR [5] and GQA [4] illustrate the value of explicit scene structure and question operations. BrickAtlas uses source structure for provenance while keeping it out of visual requests. A separate text-graph challenge tests component counting. The expressivity analysis of neighborhood aggregation against Weisfeiler-Lehman refinement [13] clarifies a limit of that challenge: our six-cycle/two-triangles pairs are indistinguishable to anonymous uniform-feature 1-WL. This is a known representational limitation instantiated by the audit, not a new impossibility result or an expectation that a simple GNN must solve the pairs.

3. Resource and Reference Contract

3.1. Preserved sources and separate roles

The historical resource has 617 questions on 24 human-authored assemblies and 15,334 placed parts in the LDraw format [7]. Original files, source transforms, questions and answers remain unchanged. Table 1 separates their measurement roles. The 140 visual parents define new observations in display-v2; the other 477 historical items remain accessible controls and structural tasks. A historical task ID is a provenance key, not a key for transferring a response into the new contract.

Visual questions receive only the declared image and text. Structural questions receive explicit graph evidence. Record-reading controls provide selected coordinates, interface records, coverage flags or author steps. Execution controls operate discrete visibility states under specified preconditions and effects. Shared instance IDs connect these artifacts for inspection; they do not make each question a perception-to-action chain.

Source bands D1–D4 describe assembly scale (Table 2). Neither part count nor low repetition of source-scoped answer IDs establishes cognitive difficulty. We use a separate table for scale rather than juxtaposing incomparable prior and size statistics on a common-looking chart.

Table 1. Historical family inventory. Stable schema key shape-match always displays as Part-type. Roles are disjoint: 140 visual, 146 structural, 290 nonconstant and 41 constant controls.

Family	Historical role	Items
Color	Core visual	73
Part-type	Core visual	67
Distance	Nonconstant control	73
Interface	Nonconstant control	87
Neighbors	Structural evidence	73
Graph deletion	Structural evidence	73
Coverage	Nonconstant control	17
Evidence limit	Constant control	24
Step lookup	Nonconstant control	55
Step sequence	Constant control	17
Restoration	Nonconstant control	58

Table 2. Source scale only. Parts count placed instances; items count historical questions. These columns have different units and are not difficulty estimates.

Source band	Sources	Parts	Items
D1	5	501	83
D2	9	2,172	175
D3	2	1,440	59
D4	8	11,221	300

3.2. Four meanings that must remain distinct

For a source hash h and instance ID i , canonical identity $c = (h, i)$ identifies the original record. Source truth $T(c)$ stores its type, color and transform. In observation o , a display mapping $d_o(\ell)$ selects the visible referent of label ℓ . The scored answer y_o is a function of the question and this observation, not necessarily of $T(c)$ alone.

Every new visual question includes the following rule: *References select objects in the supplied image only. A label such as B0001 is an image-local display reference, not a request to retrieve a historical source instance. Renders may have edited appearances or label assignments. Answer only from the current image.* This rule is part of the hashed model-visible text on both sides. The evaluator retains c , $T(c)$ and the display mapping outside the request.

Color asks, “What color is the object currently marked B0035 in this image?” Part-type asks, “Which displayed candidate label currently marks an object with the same visible part geometry as the target marked B0035? Ignore color and pose.” The complete Part-type prompt also repeats that the current image, rather than historical identities, determines the answer. Options have explicit choice IDs. The response contract requires exactly one JSON object containing a listed `choiceId`.

3.3. What is physically preserved

Source geometry and poses are immutable. An isolated or operand-only view hides other pieces, possibly including supports; it is an observation of an original assembly, not a proposed freestanding configuration. Color edits change a rendering parameter and do not rebuild an assembly. Panel controls separately show source parts and make no claim that the panel layout is an assembly.

The supplied connector graph covers recognized relations, not every contact or stability condition. The historical inventory still contains 111 other flagged tasks and 573 intersection candidates requiring human decisions. A candidate may be intended mating, an approximate mesh intersection or a source defect. This revision does not resolve those physical judgments through visual benchmark code.

4. Interventions and Exact Observations

4.1. Color: controlled rendered appearance

The 73 Color pairs use the same original geometry, source pose, camera, scale, label projections, visibility and material settings. Only the body-color parameter changes between two distinct named opaque colors. Both arms explicitly permit edited appearance; neither arm asks for the original source color. The 146 observations use Red, Blue, Yellow and Black, with aggregate counts 37, 37, 36 and 36 respectively. These are displayed gold categories, not a sample of natural material colors.

The former donor mechanism cannot support this same claim. A source inventory finds same-type, different-color donors for only 10/73 parents and identical source linear transforms for only 3/73. Replacing a donor would usually change geometry or pose together with color. The new render intervention avoids that ambiguity by retaining the original geometry in both arms. It tests named rendered appearance, not illumination invariance, material recognition or recovery of natural object color.

The machine audit checks common camera and label metadata and compares the two native images. All 73 pairs have zero changed pixels outside the common projected body mask dilated by one antialiasing pixel. Independent regeneration of three canary observations is byte-identical. These checks locate the pixel change; they cannot establish that humans perceive every chosen category unambiguously.

4.2. Part-type: candidate-reference reassignment

The 67 Part-type pairs retain the target geometry and token. The matching candidate exchanges labels with one other candidate, so the current reference that selects matching geometry changes. The option list remains fixed within the pair. Source part numbers independently derive a candidate gold, but the question asks for visible geometry. Similar

types, hidden features or a confusing label leader can defeat visual uniqueness even when the record-level match is exact.

The new question is paired with the preserved A/B image pixels under fresh observation IDs and hashes. Keeping an image unchanged does not preserve its review validity after a question changes. In the illustrated pair, B0035 remains the target, while the matching candidate's visible label changes from B0036 to B0196. The source instance that originally carried B0036 remains the same source instance. The scored answer follows its current displayed label.

A separate set of 67 position-reference observations shows a target and four candidate panels without text inside the image. Options refer to candidate positions. This provides a geometry-matching comparison under a different reference interface. Panel layout and visual access also change; a difference from label-based performance cannot by itself identify an OCR or label-tracking effect. The control has its own human reviews and never enters primary paired Both.

4.3. Freezing the input boundary

An exact observation includes the complete system/user messages, question, options, native 1280×800 PNG bytes, resolution and image-detail policy. All 347 observations are frozen before review: 280 primary endpoints and 67 auxiliary controls. Rendering metadata, canonical source truth, conditions, pairing and gold are evaluator-only records. Neither endpoint is supplied as conversational history to the other.

The reviewed observation is distinct from the provider transport envelope. A run pins its model revision, endpoint adapter and generation settings, and separately hashes the full HTTP request. The runner must reproduce the reviewed messages and image bytes inside that envelope. This supports comparisons across pinned models without asserting that a provider's internal image preprocessing is observable. Any change to reviewed text, pixels, dimensions or image policy requires new review; changing only the pinned provider envelope does not create a new visual question.

5. Human Qualification and Response Analysis

5.1. Independent review of both sides

Each of two independent reviewers receives all 347 observations in a separately shuffled queue with opaque unit IDs. The interface shows one observation at a time, its exact text and scrollable native pixels. It withholds the condition name, parent/pair membership, source record, gold, model output and the other reviewer's decision. The primary denominator is always 280; it cannot become a B-only queue through task-ID deduplication.

A raw review contains the reviewer pseudonym, times-

tamp, categorical decision, visual choice, reason, and attestations of native-size inspection and independence. Four structured checks ask whether the target reference is unique, exactly one option is visually correct, visibility is sufficient, and the observation lacks reference contradictions. Decisions distinguish decidable, ambiguous, unanswerable and contradictory cases.

Matching decisions, choices and checks produce a final judgment. Any disagreement among these fields requires explicit adjudication by a third distinct reviewer. Both original records remain intact. A final decidable choice that conflicts with the generated gold blocks finalization and requires revising the item and repeating QA, rather than silently replacing the answer. Agreement reports include raw decision agreement, multiclass Cohen’s kappa, choice/check agreement and their actual denominators. Undefined kappa remains undefined.

The runner requires final decisions for every planned endpoint before real visual inference. Nondecidable final judgments remain in the full denominator; only pairs with two final decidable endpoints enter the additional QA subset. This prevents post-result item selection. The gate recomputes decisions from retained raw files. It verifies recorded provenance and attestations, not a person’s identity through cryptography. Coordinator oversight remains necessary.

5.2. Correct adaptation and failure denominators

For pair i , let p_{Ai}, p_{Bi} be normalized choices, $y_{Ai} \neq y_{Bi}$ their golds, and v_{Ai}, v_{Bi} indicate valid output. Invalid or missing predictions are never equal to a gold. For N fixed pairs, we report

$$AAcc = N^{-1} \sum_i [p_{Ai} = y_{Ai}], \quad (1)$$

$$NewAcc = N^{-1} \sum_i [p_{Bi} = y_{Bi}], \quad (2)$$

$$Both = N^{-1} \sum_i [p_{Ai} = y_{Ai}] [p_{Bi} = y_{Bi}]. \quad (3)$$

The B outcomes partition into new-gold correctness, old-gold retention, another valid choice and invalid output. In particular,

$$R_{old} = N^{-1} \sum_i v_{Bi} [p_{Bi} = y_{Ai}]. \quad (4)$$

Old-gold retention is a behavioral outcome, not proof of memory or an internal causal mechanism.

Conditional adaptation is Both divided by A accuracy for the micro estimate. Its denominator is the number of A-correct pairs; a zero denominator yields null. Conditional source-macro analysis averages within-source conditional rates only over sources with eligible pairs and reports that

coverage. We retain unconditional estimates so conditioning cannot hide a weak A endpoint.

Format-valid rates are reported for A, B and both endpoints, alongside semantic rates. Timeouts, API failures, refusals, missing receipts and illegal formats remain incorrect in the planned denominator, with separate failure reasons. A stable invalid string is not semantic consistency. The auxiliary position control reports its own accuracy and validity without being pooled with either visual family.

5.3. Aggregation, selection and comparisons

For a pair outcome z_{si} from source s , the primary estimator is

$$\hat{\mu} = \frac{1}{|S|} \sum_{s \in S} \frac{1}{n_s} \sum_{i=1}^{n_s} z_{si}. \quad (5)$$

Each family has one primary pair per parent. We also report micro rates, pair counts and source counts. A family absent from a source does not contribute an artificial zero. Percentile intervals use 10,000 whole-source bootstrap draws with seed 20260920. Resampling sources preserves within-source dependence but does not remove dependence caused by shared parts or authoring conventions.

Full and both-decidable estimates are published together with exclusion reasons and source coverage. The QA-minus-full difference resamples the same source weights for both estimates, recomputing each mean on its eligible sources. This retains changes in source composition. Overlapping exclusion reasons are not added as independent counts. Empty subsets and bootstrap draws without selected sources are explicitly recorded.

Model comparisons use paired outcomes on the same items. Ordinary-A, NewAcc and Both ranks use exact rational point comparisons and descending midranks for ties. A reversal means the signs of the A and Both differences disagree; a tie is separately classified. Paired source intervals accompany the differences. We also record whether comparison directions agree between full and QA subsets. The intervals are descriptive and unadjusted for multiple comparisons; a point reversal alone is not a significance claim.

5.4. What would establish diagnostic value?

A useful empirical result would reveal a stable distinction ordinary accuracy obscures: for example, comparable A accuracy but different Both, or a high B old-gold retention rate despite valid formatting. The implementation produces family-by-model-by-condition outcomes, rank comparisons and traceable cases. It selects the first hash-ordered QA-decidable example of each requested transition: A-correct to new gold, A-correct to old gold, and the same valid wrong choice on both sides. Each case links exact requests, output text and raw receipts. Unavailable case types remain absent.

Table 3. Current primary visual inventory. QA final counts completed human decisions, not rendered images. No completed model runs are registered. Both-decidable coverage is 0/140 pairs; this is an evidence-availability count, not model accuracy.

Family	Pairs	Sources	QA final
Color	73	24	0/146
Part-type	67	24	0/134

This is a falsifiable expectation. If Both merely reproduces ordinary accuracy rankings, or differences mostly reflect format/API failures, the protocol’s added diagnostic value would be limited. Shared IDs and categorized errors do not locate a model’s internal failure. The present report contains no learned-model rankings, reversals or response cases. Table 3 gives the actual qualification state rather than filling a model table with synthetic scores.

6. Graph Audit: Which Controls Resist Shallow Rules?

6.1. Natural evidence and hypothetical partitions

A historical Graph deletion question asks for the number of connected components after deleting a specified node from an explicit undirected graph, including isolated survivors. Duplicate edges are deduplicated. Independent BFS and union-find implementations agree on all golds. These algorithms validate the supplied graph contract, not the completeness of the physical connector detector.

The audit retains all 73 natural questions from 24 sources. It also reclassifies the existing 219 hypothetical observations: each parent has an anchor, an answer-changing partner and an isomorphic answer-preserving partner. The 12 strong change pairs span eight sources and match surviving node set, edge count, target degree, complete surviving degree multiset and isolate count. The other 61 change pairs span 23 sources and are explicitly called *degree-visible controls*. Their invariance pairs are reported separately. The shared anchors mean these four partitions cannot be summed as independent observations.

Strong pairs compare a six-cycle with two triangles, with identical extra isolated nodes. Weak pairs use a path versus a triangle plus isolates. Both templates can match node and edge counts, but only the former matches degrees. These are hypothetical edge lists derived for a controlled test; they are not claims about newly discovered physical connections in the source assemblies.

6.2. Baselines and matching limits

We include node-count, edge-count, degree-histogram and anonymous 1-WL feature lookups; the isolates-plus-one rule; and BFS/union-find. For a surviving graph G' , the

heuristic is

$$\hat{c}_{\text{iso}} = n_0 + \mathbf{1}[|V(G')| > n_0], \quad (6)$$

where n_0 is its isolate count. It does not traverse components. Feature lookups use labels only from other sources, select the most frequent answer, break ties toward the smaller integer, and abstain on unseen features. Abstentions count as incorrect. Training uses changed endpoints once; invariance copies do not duplicate training labels. Natural and hypothetical fits remain separate.

Anonymous 1-WL initializes every surviving node identically, refines by neighbor-label multisets and compares the resulting graph signature. All strong change pairs collide under these features. Any deterministic predictor restricted to the common signature must give the same answer to two observations with different golds, making Both zero. This limitation also applies to anonymous uniform-feature message passing no more expressive than 1-WL [13]. Unique node-ID features, positional encodings, spectral or higher-order methods are outside the claim. A trained GNN experiment is not substituted by a claim that such a GNN ought to succeed.

6.3. Measured results and uncertainty

Table 4 shows that isolates-plus-one obtains 72/73 natural correct answers, compared with 73/73 for BFS. Its natural source-macro accuracy is 97.92%, with a descriptive 95% source interval [93.75, 100]. Degree-histogram and 1-WL lookups reach 69/73. These high natural scores leave little room for an ordinary accuracy comparison to distinguish component traversal from shallow features.

On degree-visible change pairs, degree-histogram lookup, 1-WL lookup and isolates-plus-one all achieve 61/61 Both. On strong change pairs, each obtains 0/12; BFS and union-find obtain 12/12. Strong invariance illustrates another distinction: the degree and 1-WL predictors are consistent across the two isomorphic observations, but only 6/12 pairs are jointly correct. Invariance of predictions alone would reward the six stable wrong pairs.

Strong answer-change directions are balanced: six increases and six decreases. Per-answer frequencies in the stored A/B arms are not identical and are fully disclosed. We use unordered Both and symmetric endpoint accuracy for the primary strong result. Giving each endpoint equal weight in both arm roles equalizes their weighted answer histograms without creating trials or inflating the 12-pair denominator. This does not make the four component-count categories uniformly distributed.

Boundary outcomes require caution. Approximate pair Wilson intervals are [0, .2425] for 0/12 and [.7575, 1] for 12/12. They treat pairs as independent although some sources repeat. Whole-source bootstrap intervals also accompany the audit, but can degenerate when every sampled

Table 4. Computed algorithmic evidence, in correct items or both-correct pairs. Natural lookups are fitted separately from hypothetical lookups, always holding out the entire test source. Strong change is the primary proof of concept; degree-visible controls cannot support the same claim. Strong invariance reuses anchors. These are algorithm outputs, not learned-model results. Source intervals and every strong pair appear in the scientific supplement and machine-readable audit.

Algorithm	Natural correct / 73	Strong change Both / 12	Degree-visible Both / 61	Strong invariance Both / 12
Node-count lookup	63	0	0	6
Edge-count lookup	44	0	0	7
Degree-histogram lookup	69	0	61	6
Anonymous 1-WL lookup	69	0	61	6
Isolates + one	72	0	61	6
BFS	73	12	61	12
Union-find	73	12	61	12

source has the same boundary outcome. Neither interval establishes population generality for eight sources and one strong construction. The claim is that these specified shallow statistics are insufficient on this proof of concept.

7. Reproducibility, Limits and Empirical Acceptance

The release preserves earlier files in content archives and records their hashes, including original source render assets. New observations receive new IDs; old prompts, reviews and responses cannot be registered as display-v2 results. The live runner saves exact requests and raw provider responses, pins the returned model identifier, reparses output and checks gold provenance before analysis. Completed live runs with one common finalized QA snapshot are the only eligible visual inputs to the reporting pipeline. Dry runs and test fixtures are explicitly excluded.

Machine verification covers 347 frozen visual requests, independently derived visual gold, the body-pixel audit, 219 graph request snapshots and 26 protocol/analysis tests. It establishes implementation behavior, not real human agreement or remote-provider authenticity. A stored receipt is auditable evidence only when the collection process and retained provider records are trustworthy.

The current release has zero genuine human reviews and zero completed visual or graph model runs. Thus the full-versus-QA comparison, model-specific retention, format effects, rankings and genuine case examples are unavailable. Those requirements remain substantive: the primary question is unresolved until both visual endpoints are independently reviewed and the new conditions are run.

The scientific supplement contains protocol details, source coverage, all 12 strong pairs, uncertainty and concrete stimulus records. The full 617 historical dossiers are a separate searchable JSON/HTML artifact, linked to the existing 3D viewer for rotation, zoom, expansion and action replay. An interactive view or paper enlargement is

not the exact input used for human qualification. Historical controls remain inspectable without being promoted into a composite measure of assembly intelligence.

The bounded scope is instance-label-grounded evidence following on two atomic visual attributes and 24 sources. Rendered colors do not cover natural color constancy; candidate-label changes do not establish general referring-expression understanding. The auxiliary panel comparison changes visual access. Graph evidence concerns a limited recognized graph and a small hypothetical challenge. Public source reuse also leaves contamination and cross-source dependence concerns. These limits define what a future positive result could establish. The present audits show why a coherent reference contract, complete human qualification and controls matched to the claimed mechanism are necessary before interpreting such a result.

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